

Brainstorming Report

Food Survey Analysis using Tableau and Flask

This document describes the brainstorming phase of the Food Survey Analysis project. The purpose of this phase was to identify the real-world problem, generate multiple solution ideas, evaluate their feasibility, and select the best approach for implementation. The project analyzes student food survey data and converts raw responses into meaningful visual insights using Tableau dashboards supported by a Flask web application.

1. Problem Statement

Educational institutions often collect food survey data containing information such as gender, GPA, weight, exercise habits, healthy feeling, favorite cuisine, and eating behavior. Since the data is stored as raw records, it is difficult to identify trends, compare categories, or understand relationships. Decision-makers require an interactive dashboard that summarizes the information clearly.

2. Project Objectives

- Analyze food habits of students.
- Study relationships between GPA, weight, and healthy feeling.
- Compare food preferences across gender.
- Visualize favorite cuisines and lifestyle choices.
- Build an interactive Tableau dashboard.
- Integrate the dashboard with a Flask application for web access.

3. Brainstorming Ideas

Different ideas were discussed during brainstorming: (1) static charts, (2) Excel dashboard, (3) Power BI dashboard, (4) Tableau dashboard with filters, (5) Flask integration for online visualization. Tableau with Flask was selected because it provides interactive dashboards, supports multiple chart types, and is easy to deploy.

4. Selected Features

Interactive Dashboard, Scatter Plot (Weight vs GPA), Histogram (GPA Distribution), Box Plot (GPA by Gender), Treemap (Favorite Cuisine), Pie Chart (Gender Distribution), Bar Chart (Total Students), Indian Food Preference Table, Healthy Feeling Analysis, Filters, Tooltips, and responsive dashboard layout.

5. Technology Stack

Python, Flask, Tableau Public/Desktop, Pandas, HTML, CSS, JavaScript, CSV Dataset, Git, GitHub, Visual Studio Code.

6. Expected Outcomes

The project enables quick analysis of food survey responses, identifies behavioral patterns, improves understanding of nutrition preferences, and demonstrates practical data analytics and dashboard development skills.

7. Feasibility Analysis

Aspect	Observation
Technical	Tableau and Flask fully support the required dashboard.
Economic	Uses open-source Python and free Tableau Public for sharing.
Operational	Simple interface with interactive filters.

Time	Project completed within the planned schedule.
Scalability	Additional survey records and dashboards can be added easily.

Conclusion

The brainstorming phase concluded that an interactive Tableau dashboard integrated with a Flask application is the most effective solution for analyzing the Food Survey dataset. The final system provides clear visualizations, supports interactive exploration, and helps users understand food preferences, health indicators, and academic relationships efficiently.